Foresight StAMP

AI Medical data Passport. A collaborator briefing on what we are submitting to SPRIND, written for everyone on the team, including those without a medical or biology background.

Why you are reading this: the four of us are applying to SPRIND, the German Federal Agency for Breakthrough Innovation, for about EUR 450,000 over 12 months. This document explains the idea in plain words, shows the two pictures we will use, reproduces the application text as it currently stands, and ends with the short list of things we need from you. Please read parts 1 and 4 at minimum.

Part 1: the idea in plain words

The problem, with an analogy

Imagine a library that has collected millions of books over decades, but the books have no titles on the spines, some are in languages nobody labelled, several were shelved in the wrong section, and nobody wrote down who is allowed to read which book. The library is enormously valuable and almost unusable. European hospitals are that library. They hold some of the richest health data in the world: DNA sequences, lab results, doctors' notes, scans. Almost none of it can be used to train AI, not because the computers are too slow, but because the data does not carry three things a machine needs: its meaning (what exactly was measured, on which scale, with which reference), its trustworthiness (who produced it, how, and whether it was checked), and its permission (which patient agreed to which use).

Today every AI project rebuilds those three things by hand, for one dataset, and then throws the work away. Patients, meanwhile, never see what their data was used for.

What Foresight StAMP is

Three things, under one name.

- The StAMP itself: a certificate, like a passport for a dataset. It has five pages, one per gate (explained below). A dataset gets a stamp on a page when it passes that gate. The StAMP can be shown to anyone without revealing the data.
- The Readiness Cell: software we install inside the hospital's own computers. It cleans, labels, and documents the data, records every step, enforces patient permissions, and lets AI models learn from the data without the data ever leaving the building. Think of it as a secure reading room inside the library: researchers can come in and read, but no book leaves.
- The StAMP Office: us, the four-person team. We bring the Cell into a hospital and run a 10-week stamping sprint with the hospital's own staff until the data passes all five gates.

The five gates

Each gate is one question anyone can understand, one test we run, and one fix if the test fails. A dataset can only attempt gate 2 after it passes gate 1, and so on.

Gate	The question	In plain words	What passing means
1 Existence	Do you know what you have?	Every file has a label, a fingerprint, and a note saying which patient sample it came from.	100 percent of files labelled and fingerprinted
2 Meaning	Does every value carry its context?	Everything uses the same map, the same dictionary, and the same ID system, so a value from one lab can be compared with a value from another.	95 percent of fields mapped to a standard vocabulary
3 Trust	Can you prove where each value came from?	Every number has a receipt: which tool, which version, which settings produced it. Suspicious samples are flagged, not silently used.	Every value traceable to its raw file and tool
4 Agency	Is permission machine-readable?	The computer, not a PDF in a drawer, knows who consented to what, until when, and can switch a patient off if they change their mind.	100 percent of records carry consent terms
5 Use	Can a model learn without the data leaving?	Outside researchers can ask questions and train models, but only get answers and models back, never the records.	A query and a training round succeed from outside

How we will prove it works

We will not wait for a hospital contract. We will run the whole method on public data first, in particular a remarkable public dataset: a single cancer patient, Sid Sijbrandij, published 39 TB of his own medical data (DNA, RNA, cell-level and tissue data, pathology slides, CT scans) from four time points and nine different laboratories. The dataset is excellent and messy in exactly the ways hospital data is messy: labs used two different reference maps for the genome, one sample's tumour-versus-healthy label was guessed rather than confirmed, one sample was contaminated with mouse cells, and no consent terms are written down anywhere. Our demo is to take this dataset through the five gates, catch every one of those problems automatically, and issue it a StAMP that states what it is and is not fit for. We add two mock examples of data a hospital has but this dataset lacks: German-language doctors' letters and structured hospital records.

What we ask SPRIND for, and what SPRIND gets

EUR 450,000 over 12 months, four people plus computing. In return SPRIND gets, as open and public infrastructure: the Readiness Cell software, the published StAMP standard (the five gates, the score, the certificate format), two certified public datasets, and a playbook for the 10-week sprint. The standard stays open; a future company can sell deployment, support and certification.

What happens if it works

Within two years, several hospitals run Cells and a national body references the StAMP in its data-quality guidance. Within ten to fifteen years, patients hold their own passport of their data, can see every model that learned from them, can revoke consent with one action, and can study their own health data, alone or in citizen-science groups they form themselves. Image analysis (scans and slides) is deliberately left for a later stage; in this project images are only registered and linked.

Who does what

Person	Background	Owns
A, Babita Singh	Biology, genomics, AI safety, data security; co-author of GA4GH Beacon v2, led the six-hospital Beacon pilot in Catalonia	Gate 2 (Meaning), standards liaison, WP1 co-lead, WP5 co-lead

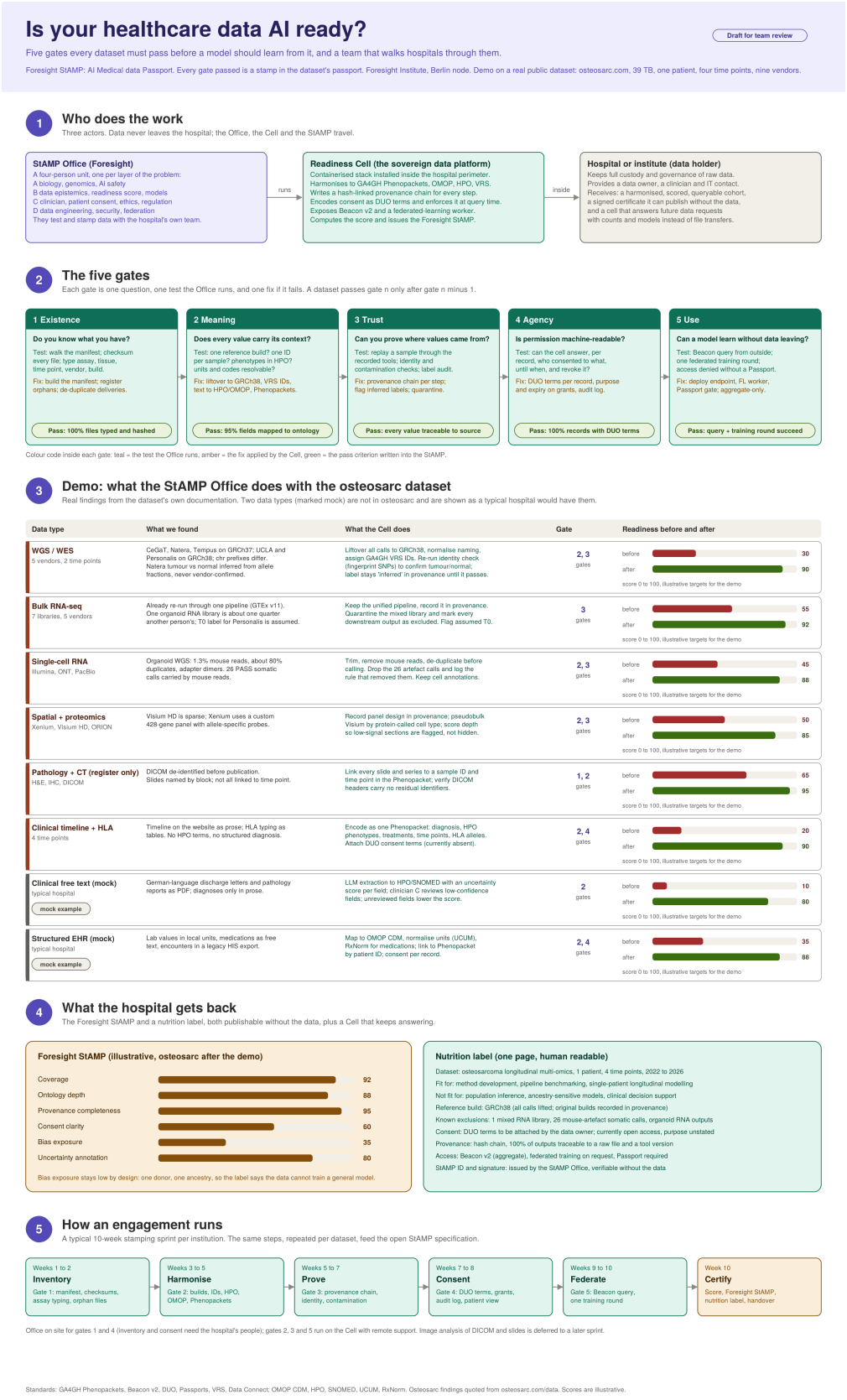
Person	Background	Owns
B	Data epistemics, in-house analysis platform, custom AI models, ethics	Gate 3 and the StAMP score, WP3 lead, WP4 co-lead
C	Medical doctor, patient-facing consent, ethics, regulation, patient advocacy	Gate 4 (Agency), WP5 lead, WP2 co-lead, clinician review
D	Data engineering, machine learning, backend and data security, privacy	Gates 1 and 5, Cell integration, WP1 co-lead, WP2 lead, WP4 co-lead

Words you will see in the application

Term	Meaning
GA4GH	Global Alliance for Genomics and Health, the body that publishes open standards for sharing genomic and health data. We build on its standards rather than inventing new ones.
Phenopacket	A GA4GH standard file that describes one patient: diagnosis, symptoms (as coded terms), genetic findings, treatments, time points.
Beacon v2	A GA4GH standard that lets someone ask a database 'do you have patients with X?' and get only a yes, no or count back, never the records.
HPO, OMOP, SNOMED	Standard dictionaries. HPO for symptoms and traits, OMOP for hospital records, SNOMED for clinical terms. Using them is what makes gate 2 possible.
DUO terms	Data Use Ontology: a GA4GH code list for consent ('research only', 'no commercial use', 'this disease only'). Gate 4 attaches these to every record.
GA4GH Passport	A credential for a person (a researcher) saying what they are allowed to access. Different from our StAMP, which is the dataset's passport.
Federated learning	Training an AI model by sending the model to each hospital's data and combining what it learned, instead of collecting the data in one place.
Provenance	The receipt trail: which raw file, which tool, which version, which settings produced each value.
Reference build (GRCh37, GRCh38)	Two versions of the standard map of the human genome. Positions differ between them, so mixing them silently corrupts analysis.

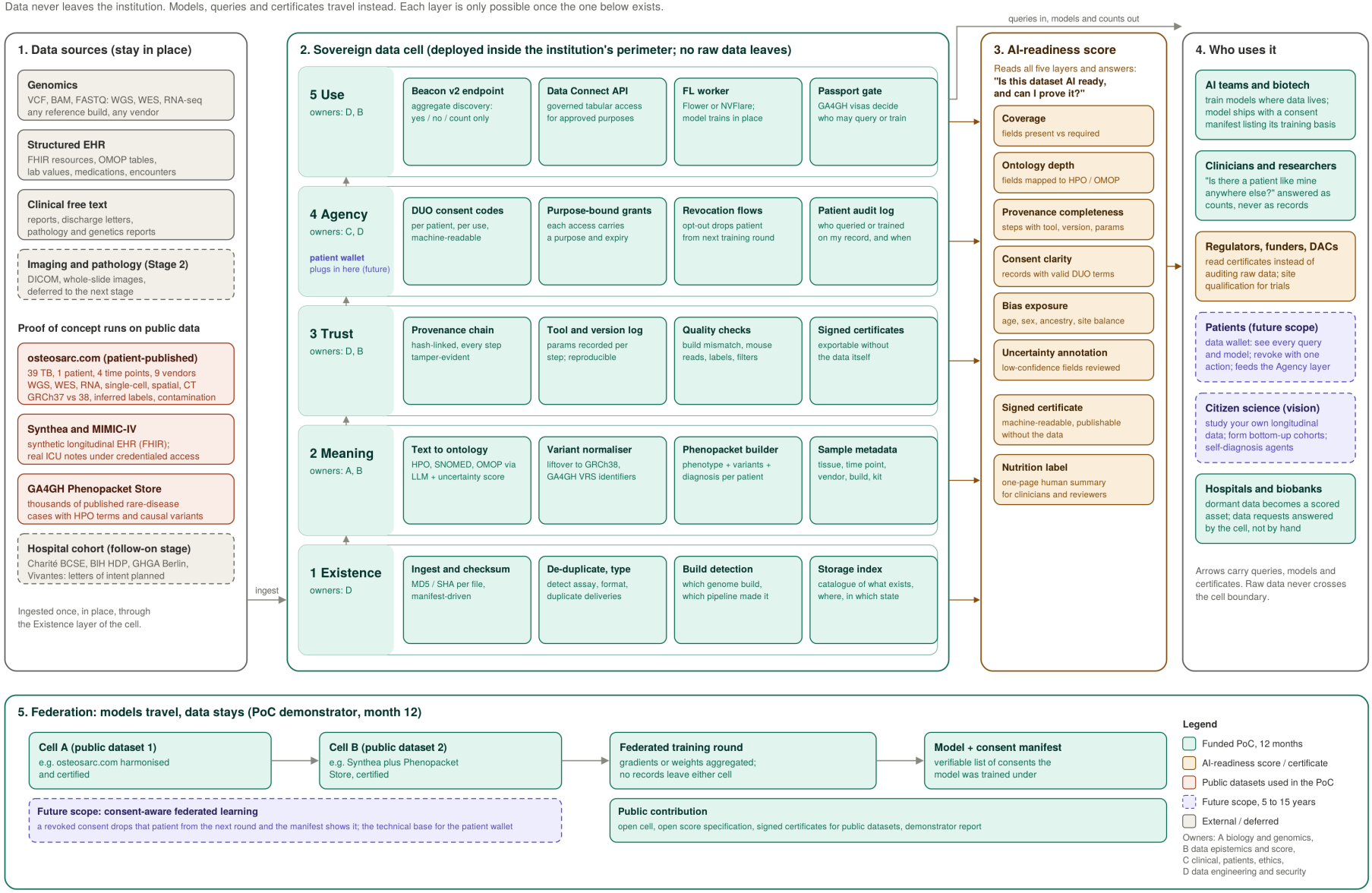
Part 2: the two pictures we will attach

The infographic is the one-page explanation for a non-specialist reviewer. The architecture schematic is for the technical reviewer.



Foresight StAMP: making medical data AI ready, sovereign and patient-owned

Data never leaves the institution. Models, queries and certificates travel instead. Each layer is only possible once the one below exists.



Standards used: GA4GH Phenopackets, Beacon v2, DUO, Passports and Visas, VRS, Data Connect; OMOP, HPO, SNOMED; ELIXIR AAI and Federated EGA for institutional deployment. Draft for team review, October 2026.

Architecture schematic: data sources, the Readiness Cell with its five layers, the StAMP score, who uses it, and the federation demonstrator.

Part 3: the application text as currently drafted

Each heading is one field of the SPRIND form; the number is its character limit including spaces. Several fields currently run over and will be trimmed before submission. Placeholders in square brackets are still to be filled. This is the general proposal route (no deadline), not the Recoding Medicine challenge.

Project name and main idea (800)

Project name: Foresight StAMP (AI Medical data Passport)

Foresight StAMP answers one question for any hospital: is your data AI ready, and can you prove it? It combines three things. A Readiness Cell, software installed inside the institution, that harmonises genomic, phenotypic and clinical-text data into open standards (GA4GH Phenopackets, OMOP, Beacon v2), records provenance, enforces patient consent and lets models train without data leaving. The StAMP itself: a five-gate certificate (Existence, Meaning, Trust, Agency, Use) that turns readiness into a measurable, signed, publishable mark. And the StAMP Office, a four-person unit that tests and stamps each hospital's data in a 10-week sprint. The proof of concept runs the full method on a real public 39 TB multi-omics dataset and releases cell, score and certificate as open infrastructure.

Challenge and goal (1500)

Europe holds some of the deepest health data in the world, and almost none of it can be used to train AI. The reasons are not technical capacity but epistemic: data exists, but it does not carry its meaning (which ontology, which assay, which reference genome), its trust (who produced it, with what quality, with which errors), or its permission (which patient consented to which use). A dataset missing any of these is not AI ready, however clean the CSV looks. Today each AI project rebuilds these layers by hand, from scratch, for one dataset, and throws the work away.

The second failure is ownership. Data is either locked inside an institution or released as a static copy that patients never see again. Neither model scales to a world where models are retrained continuously and patients expect to know what their data did.

Goal: deliver, within 12 months, (1) an open reference implementation of a sovereign data cell that makes a hospital's genomic and clinical data AI ready in place, (2) a published AI-readiness standard and scoring engine that certifies any medical dataset on five layers (existence, meaning, trust, agency, use), and (3) a demonstrator on public data proving that a model can be trained across cells by federated learning with every data point traceable to a consent. The longer goal, beyond this project, is a patient-held data wallet and consent-aware federated learning that together make citizen science on one's own health possible.

How the technology works (1500)

The method is five gates. Each gate is one question a data operator can answer, one test the Stewards run, one fix the Readiness Cell applies, and one pass criterion written into the certificate. A dataset passes gate n only after gate n minus 1. The Cell is a containerised stack deployed inside the institution's perimeter; raw inputs (VCF, BAM, FASTQ, EHR exports, free-text reports, lab tables) never leave the host.

Gate 1, Existence: do you know what you have? Test: walk the manifest, checksum every file, type assay, tissue, time point, vendor, reference build. Fix: build the manifest, register orphans, de-duplicate. Pass: 100 percent of files typed and hashed.

Gate 2, Meaning: does every value carry its context? Test: one reference build, one ID per sample, phenotypes in HPO, units and codes resolvable. Fix: liftover to GRCh38, GA4GH VRS identifiers,

LLM-assisted text to HPO, SNOMED and OMOP with an uncertainty score per field, one Phenopacket per patient. Pass: 95 percent of fields mapped to an ontology.

Gate 3, Trust: can you prove where each value came from? Test: replay a sample through the recorded tools; identity (fingerprint SNPs) and contamination checks; audit of inferred labels. Fix: a hash-linked provenance chain per transformation with tool, version and parameters; inferred labels flagged; failing data quarantined, not deleted. Pass: every value traceable to a raw file and a tool version.

Gate 4, Agency: is permission machine-readable? Test: can the Cell answer, per record, who consented to what, until when, and revoke it? Fix: GA4GH DUO terms per record, purpose and expiry on every grant, an access log the patient can read; revocation propagates to the next training round. Pass: 100 percent of records carry DUO terms.

Gate 5, Use: can a model learn without the data leaving? Test: a Beacon v2 query from outside, one federated training round, access denied without a GA4GH Passport. Fix: deploy the Beacon endpoint (aggregate answers only), a federated-learning worker (Flower or NVFlare) and the Passport gate. Pass: query and training round succeed.

The AI-readiness score reads all five gates and emits a signed, machine-readable certificate (coverage, ontology depth, provenance completeness, consent clarity, bias exposure, uncertainty annotation) plus a one-page nutrition label stating what the dataset is and is not fit for. Both can be published without the data. Imaging (DICOM, whole-slide) is registered at gate 1 and linked to samples at gate 2 in this project; image analysis itself is deferred to a later stage.

Unique selling point (1500)

Three things set Foresight StAMP apart, and one method ties them together: five gates that any data operator can understand, each with a test, a fix and a pass criterion, delivered as a repeatable 10-week sprint rather than a one-off research project.

First, it treats AI readiness as an epistemic property, not a formatting task. Existing tools clean and convert; none of them certify meaning, provenance and consent together. The five-layer score is the first auditable definition of "AI ready" for medical data, and it is designed to be adopted as a standard rather than as a product feature.

Second, it is built entirely on open GA4GH and ELIXIR standards (Phenopackets, Beacon v2, DUO, Passports, VRS, Data Connect), which already have institutional buy-in across Europe. Our team co-authored the Beacon v2 specification and ran the first multi-hospital Beacon pilot (six Catalan hospitals), so the cell is not a new silo but a packaging of the standards the community already agreed on.

Third, patient agency is enforced inside the machinery, not in a policy document. Consent travels with the data as code, revocation reaches into the federated training loop, and every model carries a verifiable manifest of the consents it was trained under. This is the technical basis for the patient data wallet and for citizen science, and no current federated-learning framework offers it.

The demonstrator strengthens the case: the cell will be run on a real, public, patient-published 39 TB longitudinal multi-omics dataset whose own documentation lists cross-vendor reference-build mismatches, inferred sample labels and contaminated libraries. Catching and certifying exactly these problems is what the score is for.

Current status and readiness (1500)

The project is at concept-plus-components stage (roughly TRL 3 for the integrated cell, TRL 5 to 6 for several parts).

Already built by the team:

- Beacon v2 specification (co-authored) and the first real-world deployment connecting six hospitals in Catalonia for rare-disease discovery; the Viral Beacon for real-time SARS-CoV-2 variant exchange across borders.
- Federated learning and clinical-text models from the GenoMed4All Horizon 2020 project: a domain-adapted BERT fine-tuned on 1,328 haematology clinical reports, used for patient stratification, plus published guidelines on federated learning with multimodal health data.
- A tamper-evident provenance chain with hash-linked certificates, built for ArtGene-Archive (biological sequence provenance), which transfers directly to data-use provenance.
- Genethropic platform components: gene-level knowledge graphs and a graph search engine across 15 species.
- [PERSON B]: in-house data analysis platform and custom model development. [PERSON D]: production data pipelines, backend security and privacy engineering. [PERSON C]: clinical practice, patient-facing consent and regulatory experience.

Not yet built: the integrated cell, the AI-readiness scoring engine, consent enforcement inside the training loop, and the certificate format. No hospital partner is signed; the proof of concept is designed to run on public data so that institutional access is not on the critical path.

Proof of concept plan on public data (1500)

The proof of concept uses three public sources chosen to cover depth, breadth and standards:

- Depth and real-world mess: the osteosarcoma dataset published by the patient himself (osteosarc.com), 39 TB across four time points, nine vendors, WGS, WES, bulk and single-cell RNA, spatial, pathology, CT and HLA. Its own documentation lists GRCh37 versus GRCh38 mismatches, inferred tumour/normal labels, mouse contamination and a mixed RNA library. The cell will ingest, harmonise and certify it, and the score must flag each documented defect.
- Breadth and clinical text: Synthea synthetic longitudinal EHRs (FHIR) and, under credentialed access, MIMIC-IV clinical notes, to exercise the text-to-ontology layer at scale.
- Standards and cohort shape: the GA4GH Phenopacket Store (thousands of published rare-disease cases with HPO terms and causal variants) as a reference cohort.

What the Stewards will do with the osteosarcoma dataset, gate by gate, using findings from its own documentation:

Data type	Found	Cell action	Gates
WGS / WES, 5 vendors	CeGaT, Natera, Tempus on GRCh37; UCLA, Personalis on GRCh38; tumour vs normal inferred from allele fractions	Liftover to GRCh38, VRS IDs, fingerprint-SNP identity check; label stays "inferred" until it passes	2, 3
Bulk RNA-seq, 7 libraries	One organoid RNA library about one quarter another person's; one T0 label assumed	Keep unified GTEx v11 pipeline in provenance; quarantine mixed library; flag assumed label	3
Single-cell RNA	1.3 percent mouse reads, about 80 percent duplicates, 26 somatic calls carried by mouse reads	Trim, remove mouse reads, de-duplicate; drop and log the 26 artefact calls	2, 3
Spatial, proteomics	Sparse Visium HD; custom 428-gene Xenium panel	Record panel design; pseudobulk by cell type; depth score flags low-signal sections	2, 3
Pathology, CT	DICOM de-identified; slides named by block, not all linked to time point	Register and link every slide and series to sample and time point; image analysis deferred	1, 2

Data type	Found	Cell action	Gates
Clinical timeline, HLA	Prose timeline, HLA tables, no HPO, no DUO terms	One Phenopacket: diagnosis, HPO, treatments, time points, HLA; DUO terms attached	2, 4

Two data types a typical hospital holds but osteosarc lacks are added as mock inputs: German clinical free text (LLM extraction to HPO/SNOMED with clinician review of low-confidence fields) and structured EHR exports (OMOP CDM, UCUM units, RxNorm).

Deliverables: two Cells on separate infrastructure, a Beacon v2 query answered across both, one federated training round with a consent manifest, and a signed certificate plus nutrition label for each dataset. The nutrition label will state plainly that a one-donor dataset is fit for method development and not for population inference.

Milestone for later stages: replace one public cell with a European hospital cell. Conversations are planned with Charité (BCSE, BIH Health Data Platform), GHGA Berlin and Vivantes; [ADVOCACY PARTNER] supports patient-side validation of the consent model.

Milestones, work packages and budget (1500)

Total request: EUR 450,000 over 12 months, four Stewards plus infrastructure. The work packages build the Cell and the score in the first half and run two full 10-week readiness sprints on public data in the second half.

WP	Scope	Lead	Months	EUR k
WP1 Readiness Cell, gates 1 and 2	Ingestion, manifest, harmonisation (HPO, OMOP, Phenopackets, VRS), Beacon v2 and Data Connect endpoints	A, D	1 to 7	115
WP2 Gates 3 and 4	Provenance chain, identity and contamination checks, DUO consent, Passport access, revocation	D, C	2 to 8	85
WP3 Score and certificate	Five-gate metric, scoring engine, signed certificate, nutrition label, open specification	B, A	3 to 10	85
WP4 Readiness sprints and federation, gate 5	Two 10-week sprints (osteosarc, Synthea plus Phenopacket Store), FL worker, consent manifest, evaluation	B, D	5 to 12	85
WP5 Patients, ethics, standards	Consent UX with [ADVOCACY PARTNER], GDPR and EHDS review, GA4GH liaison, hospital letters of intent, open release	C, A	1 to 12	50
Infra and ops	Compute and storage for the 40 TB demonstrator, security review, travel	all	1 to 12	30

Milestones: M3 gates 1 and 2 pass on Synthea and Phenopacket Store; M6 Cell v1 with provenance and consent, osteosarc sprint started; M8 first signed certificates; M10 second sprint complete, both datasets certified; M12 two-Cell federated training with consent manifest, public release of Cell, score specification, sprint playbook and demonstrator report.

Personnel is about 80 percent of cost (four people at roughly 0.8 to 1.0 FTE); the remainder is compute, storage, security audit and dissemination.

Financing so far and why SPRIND (1500)

The components were financed through earlier public research grants (Horizon 2020 GenoMed4All, ELIXIR and EGA/CRG institutional funding for Beacon), hackathon prizes and the founders' own time through Genethropic. No external money has been raised for Foresight StAMP itself.

Why other sources do not fit: venture capital will not fund an open standard and reference implementation whose value accrues to hospitals and patients rather than to a single vendor; classic academic grants fund papers, not a deployable, security-reviewed product; and EU consortia (Horizon Europe, EHDS pilots) take 12 to 18 months to assemble and reward large incumbents over a four-person team with a focused idea. The Recoding Medicine challenge is adjacent but requires a signed dataset at submission, which we deliberately avoid by building on public data first so the infrastructure is not held hostage to one institution's data agreement.

SPRIND's model (fast, milestone-based, tolerant of standards work with delayed commercial returns, with regulatory and partner coaching) is the one instrument that fits a project whose first output is public infrastructure and whose commercial layer comes second.

Definition of success (1500)

During the project:

- Two cells run on independent infrastructure and answer a joint Beacon v2 query at month 12.
- The osteosarcoma dataset is fully ingested and certified, and the score flags every defect documented by its publishers (reference-build mismatches, inferred labels, contamination) without being told about them.
- A model trained by federated learning across the two cells carries a verifiable consent manifest; revoking one synthetic patient's consent removes them from the next training round and the manifest shows it.
- The AI-readiness score specification, the cell and the scoring engine are released under an open licence with documentation good enough for an external team to deploy a cell in under a week.
- At least one German or Spanish hospital has signed a letter of intent to host a cell in a follow-on stage.

After completion:

- Three or more institutions run cells within 24 months and at least one national body (GHGA, Medical Informatics Initiative, genomDE, ELIXIR node) references the score in its data-quality guidance.
- A commercial entity offers the cell as a supported product with a sustainable revenue line, while the standard stays open.
- The patient wallet pilot starts with [ADVOCACY PARTNER], with real patients seeing for the first time which models their data trained.

Challenges and risks (1500)

- Standards drift: GA4GH and OMOP evolve; the cell must track versions without breaking certificates. Mitigation: pin versions in the provenance chain, treat migration as a scored event.
- Harmonisation quality: LLM-assisted mapping of clinical text to ontologies can be confidently wrong. Mitigation: uncertainty scores per field, human-in-the-loop review by person C, and the score penalises unreviewed low-confidence fields.
- Scope creep into imaging and wearables. Mitigation: genomics plus phenotype plus clinical text only in this stage; imaging deferred.
- No hospital partner at start. Mitigation: public data proves the stack; institutional cells are a follow-on milestone, with conversations started early.
- Legal: GDPR, the European Health Data Space and national rules on secondary use differ; consent-as-code must be reviewed. Mitigation: ethics and regulation owned by person C, external

legal review budgeted.

- Adoption: hospitals have little incentive to make data AI ready. Mitigation: the certificate gives them a visible asset (a scored dataset) and the cell lowers their cost of answering data requests.
- Security: a cell is an attractive target. Mitigation: no raw data leaves the perimeter, aggregate-only Beacon responses, Passport-based access, external security review in the budget.
- Team: four people, cross-border. Mitigation: clear WP ownership and a single integrator (person D).

How it is done today and why it fails (1500)

Today, making medical data AI ready is a bespoke project repeated for every study. A data scientist requests an export, waits months for a data access committee, receives a CSV or VCF with undocumented conventions, spends most of the project cleaning and mapping it, trains a model, and discards the cleaning work. Consent is handled as a PDF attached to the request, not as something the system enforces. Provenance lives in a notebook, if anywhere.

Large initiatives have improved parts of this. The Medical Informatics Initiative and EHDS define common data models; GA4GH and ELIXIR define standards for discovery and access; federated-learning frameworks let models train in place. But each solves one layer: formats without meaning, access without consent enforcement, training without provenance. No existing tool answers the question "is this dataset AI ready, and can I prove it?" with a certificate, and none hands the patient a view of what their data did.

The result is that Europe's richest datasets stay dormant, AI teams work on whatever is easiest to obtain rather than what is clinically valuable, and patients are asked to trust a process they cannot see. The problem has not been solved because the fix sits between disciplines: it needs genomics, software engineering, clinical practice and ethics in one team, which is exactly how the StAMP Office is staffed.

Competitors and similar projects (1500)

Who	What they do	Similar	Different
GA4GH, ELIXIR, Federated EGA, GHGA	Standards and archives for genomic data sharing	We build on their specs	They do not score readiness, enforce consent in training, or package a deployable cell; we are a complement, not a rival
Medical Informatics Initiative, EHDS, genomDE	National and EU data infrastructure	Common data models	Top-down, slow; no patient-facing agency; no certificate
Owkin, Rhino Health, Apheris, Lifebit	Commercial federated learning and trusted research environments	Models train in place	Closed platforms, vendor lock-in, no open readiness standard, consent as policy not code
Flower, NVFlare, OpenFL	Open federated-learning frameworks	We use one as the worker	No data harmonisation, provenance or consent manifest
OHDSI, Phenopacket tooling, OpenEBench	Harmonisation and benchmarking	Ontologies and benchmarks	Single-layer tools; no integrated cell, no certificate
Matchmaker Exchange, RD-Connect GPAP	Rare-disease case matching	Federated discovery	Narrow to case matching; not a general readiness layer
Data-trust and wallet startups (various)	Patient-held health records	Patient agency	Mostly consumer apps without genomic depth or institutional deployment

Foresight StAMP is the only approach that combines an open, auditable definition of AI readiness with a deployable cell and consent enforced inside the learning loop.

Beneficiaries and customers (1500)

Beneficiaries:

- Patients, especially those with rare diseases and cancers, whose data becomes usable for models that may change their diagnosis, and who gain a view of what their data did.
- Hospitals and research institutes, which turn dormant data into a scored, reusable asset and cut the cost of each data request.
- AI teams and biotech, which get certified, harmonised data without rebuilding the cleaning layer per project.
- Regulators and funders, which get a measurable standard for data quality and consent to point to.

Customers (post-project):

- Hospital networks and university medical centres: a supported cell deployment with maintenance and certification (subscription).
- National data bodies and biobanks: certification services and a hosted scoring engine.
- Pharma and AI companies: paid, consent-respecting federated access to certified cohorts, with the data holder and patients credited.
- Clinical trial sponsors: readiness certificates as part of site qualification.

The open standard and reference implementation remain free; revenue comes from deployment, support, certification and brokered federated access.

The world if fully realised (800)

Every hospital and biobank in Europe runs a cell. Any dataset carries a readiness certificate the way food carries a nutrition label, and models are trained only on data whose meaning, provenance and consent are machine-verified. Patients hold a wallet showing which models learned from them, can revoke consent with one action, and can run their own analyses on their own longitudinal data, alone or in citizen-science cohorts they form themselves. Rare-disease diagnosis no longer depends on which hospital you walked into. Medical AI is trained on sovereign data by default, and data has become a currency that patients and institutions, not platforms, control.

Team structure and gaps (800)

The StAMP Office: four people, one per gate of the problem, hosted at the Foresight Institute, Berlin node. The Office both builds the Cell and run the readiness sprints with hospitals, on site for gates 1 and 4 (inventory and consent need the hospital's people) and remotely for gates 2, 3 and 5.

- [PERSON A], PhD: genomics, cancer and rare-disease biology, AI safety and data security; co-author of Beacon v2, Viral Beacon, GenoMed4All; founder of Genethropic. Owns Meaning and the standards liaison.
- [PERSON B]: data epistemics, in-house analysis platform developer, custom AI models, ethics. Owns the AI-readiness score.
- [PERSON C], MD: clinical practice, patient data safety, ethics and regulation, patient advocacy. Owns Agency and the patient partner.
- [PERSON D]: data engineering, machine learning, backend and data security, privacy. Owns the cell integration and Use.

Gaps: a legal specialist for GDPR and EHDS review, and a hospital-side clinical informatics contact; both planned as contracted or partner roles.

How SPRIND can support beyond money (800)

Three things matter more than funding. First, introductions to German data holders: Charité and BIH, GHGA, Medical Informatics Initiative sites and the genomDE model project, so the first institutional cell is in Germany. Second, regulatory coaching on GDPR, EHDS and the secondary-use rules, and a path to having the readiness score referenced by a national body such as BfArM or GHGA. Third, visibility: a place at SPRIND demo days with hospital CIOs, insurers and investors, so the cell can move from open infrastructure to a supported product with a commercial entity behind it.

Further remarks, attachments and open items

Further remarks (500): the team is distributed across Berlin and Barcelona; the applying entity is the Foresight Institute's Berlin node, which also hosts the Recoding Medicine matchmaking platform. All code and the score specification will be released under an open licence; a separate commercial entity is planned for deployment and support.

Attachments to prepare (PDF or TXT, 15 MB total):

- One-page concept note with the pyramid and architecture schematics
- CVs of the four team members
- Beacon v2 paper and six-hospital pilot slides
- GenoMed4All paper and federated-learning guidelines
- Letter of partnership from [ADVOCACY PARTNER]
- Short note on the osteosarcoma demonstrator plan

Names used in this draft: Foresight StAMP (the project and the certificate, AI Medical data Passport; each gate passed is a stamp in the dataset's passport), Readiness Cell (the software installed at the hospital), nutrition label (the human-readable summary), StAMP Office (the team), stamping sprint (the 10-week engagement). Image analysis of DICOM and whole-slide data is out of scope for this sprint; images are only registered and linked.

Open items for the team:

- Pick the project name
- Confirm legal form and signatory at Foresight Institute Berlin
- Confirm WP leads and FTE per person
- Decide whether MIMIC credentialing is worth the lead time or Synthea alone is enough for Stage 1
- Name the advocacy partner
- Decide which form fields need trimming to the character limits

Part 4: what we need from you

Everything about scope, roles and budget is agreed. Only Foresight Institute appears as an entity; there will be no letters from hospitals or other organisations at submission. What remains is small and physical:

Who	What	Format
B, C, D (each)	Your CV	PDF, one to two pages; all attachments together must stay under 15 MB
B, C, D (each)	Full name, title, current affiliation and position, as you want them printed in the team section	One line by email
B, C, D (each)	A one-sentence confirmation that you have no conflict of interest with SPRIND or its jury	One line by email
C	The name of the patient advocacy organisation you are in contact with, if you want it mentioned as a contact (not as a partner)	One line by email
D	Is EUR 30,000 enough for computing and storage for the demonstrator (about 40 TB) for 12 months? Yes, or a revised figure	One line by email
Foresight (via Babita)	Legal form, registered address, and the name of the person who submits	For the applicant fields of the form
Babita	Final decision on route (general proposal vs challenge), trim to character limits, own CV, declaration answers	Before submission

If you disagree with anything in parts 1 to 3, say so now; after submission the text is fixed. Comments on the shared draft document or by email are both fine.